

Taha Afzal

✉ tahaafzal5@hotmail.com ☎ (224) 522-3432 🌐 tahaafzal5 🔄 tahaafzal5 🔗 tahaafzal.com

WORK EXPERIENCE

- Software Engineer II, CNH** Dec 2024 – Present | Sioux Falls, SD
- Designing and implementing features connecting user-facing functionality with vehicle subsystems using Qt and C++ in an Agile team.
 - Developing robust C++ interfaces that integrate the UI with underlying system components and enable reliable system functionality.
 - Debugging multi-threaded software using gdb-multiarch, analyzing system logs, and documenting code with Doxygen.
 - Improving release cycles by 20% by implementing and maintaining CI/CD pipelines, automating build, test, and release workflows.
- Software Engineer, CNH** May 2022 – Nov 2024 | Sioux Falls, SD
- Collaborated with the team to transition the Viper 4+ build system from qmake to CMake, reducing build times by 25%.
 - Implemented features to create field boundaries, ensuring efficient chemical usage and preventing accidental sprays in waterways.
 - Enhanced a file conversion library, expanding support to JSON, CSV, and XML formats, ensuring 99.9% error-free conversions.
 - Mentored an intern in implementing the Delta-T quality measurement system to assess spray effectiveness based on weather conditions.
- Software Engineer Intern, Raven Industries, Inc.** Sep 2021 – May 2022 | Sioux Falls, SD
- Implemented on-device creation of geospatial vector data format shapefiles for the multiline feature in Raven Operating Software.
 - Developed this highly-requested feature for the Field Computer product line using Qt and C++ in an Agile environment.
 - Improved the multiline feature to make it more user-friendly and easier to discover enabling farmers to save 60% of their time.
- Software Engineer Intern, Raven Industries, Inc.** May 2021 – Aug 2021 | Sioux Falls, SD
- Collaborated in an Agile team of 8 to implement single sign-on for all 4 customer-facing apps for a secure user-authorization process.
 - Undertook displaying telemetry and generating alerts with Grafana to monitor load and throughput of critical services more closely.
 - Streamlined infrastructure with Terraform, reduced moving parts, refactored code, and improved documentation.
- Software Engineer Intern, Spidreye** Feb 2020 – Jun 2020 | New York, NY
- Developed, deployed, and tested a property-review website that gained 75+ users in the first month after launch in a team of 4.
 - Implemented Google Maps API to support location auto-completion when a user is searching for properties for faster results.
 - Utilized Firebase Authentication for user profiles and sending automated welcome emails through Mailchimp.

SKILLS

C++ | Python | Rust | Swift | JavaScript | SQL | Bash | ReactJS | Qt | QML | Terraform | AWS | Docker | Git | Java

EDUCATION

- M.S. Computer Science**, Georgia Institute of Technology Jan 2026 – Present
- B.A. Computer Science and Software Engineering**, Augustana University Aug 2018 – May 2022 | GPA: 3.76
- Honors and Awards**
- Global Leader Scholar, Slate Ronning International Scholar, Fred C. and J. Marie Christopherson Scholar, AURAFA Award Recipient

PROJECTS

- Uni Go, Co-Founder** Oct 2023 – Present
- Co-founded the re-branded development of Ridesio 🚗 to bring an affordable, and communal mode of transportation to college campuses.
 - Partnering with 3 college campuses to integrate Uni Go, delivering a tailored solution that meets the unique needs of each college.
- Health Memo 📄, Tech Lead** Oct 2020 – Nov 2020
- Led a team of 3 to develop an iOS app to help the elderly manage their chronic diseases and share regular updates with their doctor.
 - Utilized EMRs and Google's Text-to-Speech and Speech-to-Text to create a personalized experience for patients.
 - Won \$9,000 along with an invitation to work with engineers at Sanford Health for further research and development of the project.

LEADERSHIP AND AFFILIATIONS

- Augustana University, Trustee** Dec 2024 – Present
- Serving as a trustee on the Board of Trustees at Augustana, focusing on the well-being of faculty, students, and staff.
 - Strengthening the Computer Science program, advancing international education, and driving entrepreneurial initiatives on campus.